

Cell Biology Basics

Structure, function, and life processes inside the cell

This material is designed as a concise student handout. It introduces core ideas, important vocabulary, and short review questions for classroom or self-study use.

Learning outcomes

- Explain the core concepts using accurate biology terminology.
- Connect biological structures to their functions.
- Apply the concepts to simple examples and review questions.
- Use the vocabulary section to strengthen scientific communication.

1. What is a cell?

A cell is the basic structural and functional unit of life. All living organisms are made of one or more cells, and every cell carries out processes that keep the organism alive.

Cells are not just tiny bags of fluid. They contain organized internal parts that help them obtain energy, build molecules, remove waste, and respond to the environment.

Key points

- Cells are the smallest units that can perform life processes.
- Unicellular organisms consist of one cell; multicellular organisms consist of many specialized cells.
- Cell function depends strongly on cell structure.

Mini activity

Draw a simple animal cell and label the nucleus, cytoplasm, cell membrane, and mitochondrion.

2. Cell membrane and transport

The cell membrane controls what enters and leaves the cell. It is selectively permeable, meaning some substances can pass through more easily than others.

Transport can be passive, such as diffusion and osmosis, or active, which requires energy to move substances against a concentration gradient.

Key points

- Diffusion moves particles from high to low concentration.
- Osmosis is diffusion of water across a selectively permeable membrane.
- Active transport requires cellular energy.

Mini activity

Place a raisin in water and observe its size before and after soaking. Relate the change to osmosis.

3. Organelles and specialization

Eukaryotic cells contain membrane-bound organelles. Each organelle performs a specific job, similar to departments in a small factory.

For example, mitochondria release usable energy, ribosomes build proteins, and the nucleus stores genetic instructions.

Key points

- Nucleus: stores DNA and controls many cell activities.
- Mitochondria: release energy through respiration.
- Ribosomes: assemble proteins.

Mini activity

Create a cell analogy. Compare each organelle to a part of a school, city, or factory.

Important vocabulary

Term	Meaning
Cell membrane	Thin boundary that controls movement into and out of the cell.
Cytoplasm	Jelly-like material where many reactions occur.
Organelle	Specialized cell structure with a specific function.
Diffusion	Movement of particles from high to low concentration.
Osmosis	Movement of water through a selectively permeable membrane.

Quick review quiz

1. What is the main function of the cell membrane?
2. How is diffusion different from active transport?
3. Why do muscle cells usually contain many mitochondria?
4. Name two organelles and explain their functions.
5. What might happen to a cell placed in a very salty solution?

Teacher note: Students should answer the review questions before checking a textbook, notes, or class discussion.